

Complemented Subspaces And Order Continuity In Free Banach Lattices

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June 13, 2026

Source And Target

Youssef Azouzi and Wassim Dhifaoui, *Order denseness in free Banach lattices*, arXiv:2508.11648.

The source paper revisits a theorem of Oikhberg–Taylor–Tradacete–Troitsky asserting that, if F is a closed subspace of E , then the inclusion-induced homomorphism

$$\bar{\iota} : \text{FBL}^{(p)}[F] \longrightarrow \text{FBL}^{(p)}[E]$$

is order continuous. The paper explains that the old proof used order denseness of $\text{FVL}[F]$ in $\text{FBL}^{(p)}[F]$, which holds only when F is finite-dimensional. It proves the statement for $p = \infty$ and says that, for other values of p , the question remains open.

This shows that $y = 0$ and consequently, $\{0, f\} \cap \text{FVL}[E] = \{0\}$ as required. Therefore, we conclude that the free vector lattice $\text{FVL}[E]$ is not order dense in $\text{FBL}^{(p)}[E]$, which completes the proof. ■

Consider two Banach spaces E and F and a bounded operator $T : F \rightarrow E$. Then as noted above we can associate to T , in a natural way, a lattice homomorphism $\bar{T} : \text{FBL}^{(p)}[F] \rightarrow \text{FBL}^{(p)}[E]$ by extending T to $\text{FBL}^{(p)}[F]$ and then compose with the natural embedding $E \hookrightarrow \text{FBL}^{(p)}[E]$. One of the main ideas studied in [13] is the relationship between operators T and $\bar{T}$. In [13] the authors showed that if $\mathcal{P}$ is one of the properties – injectivity, surjectivity and bijectivity – then T has property $\mathcal{P}$ iff $\bar{T}$ does. Among these, only the proof of injectivity requires order denseness. For this reason we provide a corrected proof for it.

Proposition 6 *Let E and F be two Banach spaces, $T \in L(F, E)$ and $\bar{T} : \text{FBL}^{(p)}[F] \rightarrow \text{FBL}^{(p)}[E]$ be the associated lattice homomorphism introduced above. Then T is injective iff $\bar{T}$ is injective.*

Proof. We need only to prove the forward implication. Assume then that T is injective and let $f \in \text{FBL}^{(p)}[F]$ such that $\bar{T}f = 0$. Then f vanishes on the range $T^*(E^*)$ of T^* . As f is bounded weak* continuous f vanishes on $\overline{T^*(E^*)}^{bw*}$. As T is injective T^* has a weak* dense range and it is sufficient to show that $\overline{T^*(E^*)}^{bw*} = \overline{T^*(E^*)}^{w*}$. But this follows from Banach-Dieudonné Theorem (Theorem 2). ■

In [13, Theorem 3.4] it was shown if F is closed subspace of E and $\iota : F \hookrightarrow E$ is the inclusion map then $\bar{\iota} : \text{FBL}^{(p)}[F] \rightarrow \text{FBL}^{(p)}[E]$ is order continuous. However, the proof relies on order denseness of $\text{FVL}[F]$ in $\text{FBL}^{(p)}[F]$ which holds only in the finite-dimensional. The converse problem was also raised (see [13, Remark 3.5]). What we have been were able to prove that the former result is true for $p = \infty$; for other values of p , the question remains open. Furthermore, we give a partial answer concerning the converse problem. Before presenting the proof of these results let us state an interesting result, which is very close to [13, Lemma 2.9].

Theorem 7 *Let U be a bounded weak* open subset of E^* and $0 \neq x_0^* \in U$. Then there exists f in $\text{FBL}^{(\infty)}[E]^+$ such that $f(x_0^*) \neq 0$ and $f = 0$ outside of cone (U) . Moreover if $h \in H_\infty[E]^+$ satisfies $h(x^*) > 1$ for all $x^* \in U$ then*

Result

Theorem 1. *Let $1 \leq p \leq \infty$. Let $T : F \rightarrow E$ be a bounded operator between Banach spaces. Suppose that T has a bounded left inverse, i.e. there is a bounded operator $R : E \rightarrow F$ such that $RT = \text{Id}_F$. Then the induced lattice homomorphism*

$$\bar{T} : \text{FBL}^{(p)}[F] \rightarrow \text{FBL}^{(p)}[E], \quad \bar{T}f = f \circ T^*,$$

is order continuous.

Corollary 1. *If F is a complemented subspace of E , then the natural inclusion-induced homomorphism*

$$\text{FBL}^{(p)}[F] \longrightarrow \text{FBL}^{(p)}[E]$$

is order continuous for every $1 \leq p \leq \infty$.

Proof

We use the standard functorial formula for free p -convex Banach lattices: if $A : X \rightarrow Y$ is bounded, then

$$\overline{A} : \text{FBL}^{(p)}[X] \rightarrow \text{FBL}^{(p)}[Y], \quad \overline{A}f = f \circ A^*.$$

Let (f_α) be a decreasing net in $\text{FBL}^{(p)}[F]_+$ with infimum 0. To prove order continuity of $\overline{T}$, suppose that

$$0 \leq h \leq \overline{T}f_\alpha \quad (\alpha)$$

for some $h \in \text{FBL}^{(p)}[E]_+$. We show that $h = 0$.

Assume otherwise. Since $\text{FBL}^{(p)}[E]$ is represented as a lattice of positively homogeneous functions on E^* , there is $x^* \in E^*$ such that $h(x^*) > 0$. Put

$$y^* = T^*x^* \in F^*.$$

Then $y^* \neq 0$, because if $y^* = 0$, then

$$h(x^*) \leq (\overline{T}f_\alpha)(x^*) = f_\alpha(0) = 0.$$

We now build a bounded operator $S : E \rightarrow E$ satisfying

$$ST = T, \quad S^*R^*y^* = x^*.$$

Set $d = x^* - R^*y^*$. For every $u \in F$,

$$d(Tu) = x^*(Tu) - y^*(RTu) = y^*(u) - y^*(u) = 0.$$

Choose $u_0 \in F$ such that $y^*(u_0) = 1$, and define

$$Se = e + d(e)Tu_0 \quad (e \in E).$$

Then S fixes $T(F)$, hence $ST = T$. Moreover, for every $e \in E$,

$$\begin{aligned} (S^*R^*y^*)(e) &= (R^*y^*)(Se) \\ &= y^*(Re + d(e)RTu_0) \\ &= (R^*y^*)(e) + d(e)y^*(u_0) \\ &= x^*(e). \end{aligned}$$

Apply the induced positive lattice homomorphisms $\overline{S}$ and $\overline{R}$. Since $ST = T$, we have $\overline{S}\overline{T} = \overline{T}$. Since $RT = \text{Id}_F$, we have $\overline{R}\overline{T} = \text{Id}$ on $\text{FBL}^{(p)}[F]$. Therefore, for every α ,

$$0 \leq \overline{R}\overline{S}h \leq \overline{R}\overline{S}\overline{T}f_\alpha = \overline{R}\overline{T}f_\alpha = f_\alpha.$$

But

$$(\overline{R}\overline{S}h)(y^*) = h(S^*R^*y^*) = h(x^*) > 0.$$

Thus $\overline{R}\overline{S}h$ is a nonzero positive element of $\text{FBL}^{(p)}[F]$ lying below every f_α , contradicting $f_\alpha \downarrow 0$. Hence $h = 0$, proving that $\overline{T}f_\alpha \downarrow 0$.

Scope And Limitations

This proves the finite- p closed-subspace order-continuity statement under the additional assumption that the subspace is complemented. Equivalently, it proves the operator version when T is an embedding with complemented range.

The proof does not settle the source paper's remaining open case for arbitrary closed subspaces. The point of the complemented hypothesis is that the left inverse R lets us pull a hypothetical positive lower bound back to $\text{FBL}^{(p)}[F]$.

Novelty And Duplicate Check

Local duplicate check: the run registry, solution index, active claims, and attempt logs were checked. No packaged result covers arXiv:2508.11648. A prior search log identified the finite- p order-continuity question as plausible but did not solve or promote any subcase.

No external literature search was performed for this partial packet. The novelty claim is therefore limited: this should be read as a compact complemented-subspace proof extracted within the run, not as a claim that the subcase is absent from the literature.

Verifier Focus

The main points to check are:

1. the orientation of the functorial identities $\overline{S}\overline{T} = \overline{T}$ and $\overline{R}\overline{T} = \text{Id}$;
2. the construction of S , especially $ST = T$ and $S^*R^*y^* = x^*$;
3. the order-continuity criterion using positive lower bounds for a decreasing net.